

Tamir Yirga

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Seattle, WA – Open to Relocation

EDUCATION

NORTHEASTERN UNIVERSITY

Seattle, WA

Master of Science in Computer Science | GPA: 4.0/4.0

Expected Dec 2026

Coursework: Scalable Distributed Systems, Machine Learning, Computer Vision, Algorithms, Advanced Software Dev

Research Assistant: Co-authored a paper linking data prevalence to LLM hallucinations, further probing activation states

Teaching Assistant: Scalable Distributed Systems, Computer Vision

EXPERIENCE

LakeSail

Jan 2026 – May 2026

Software Engineer Intern

San Francisco, CA

- Engineered SQL time types and UDF/UDTF runtimes to achieve Spark 4.1.1 parity, collaborating with core maintainers to merge 4 major PRs into the Rust-native query engine
- Implemented an Arrow-native vectorized runtime for Python UDFs/UDTFs, eliminating serialization overhead to deliver a 15x throughput improvement
- Created a benchmarking suite to compare performance against JVM-based Spark across diverse ML/AI workloads, validating core use cases and identifying engine optimization limits

Canaria

Jul 2025 – Sep 2025

Software Engineer Intern

New York, NY

- Architected a FastAPI RAG API for 700M+ records, using materialized views to execute PostGIS and time-series queries 50x faster than standard JOINS
- Designed a 3-stage GPT-4 pipeline with dynamic 28-column schema injection over a rolling 30-day window, achieving 98% text-to-SQL accuracy
- Implemented Redis query caching to bypass expensive LLM calls, cutting repeat text-to-SQL latency from 4s to <200ms

Scandiweb

May 2022 – Aug 2024

Full Stack Engineer

Riga, Latvia

- Engineered React/PHP features for 8+ platforms (\$50M+ revenue); rebuilt checkouts to cut cart abandonment by 18%
- Refactored 60% of legacy services across 15+ custom extensions into service-oriented architectures with zero downtime

Endubis Labs

Apr 2021 – May 2022

Full Stack Engineer

Addis Ababa, Ethiopia

- Built a secure wallet for 7K+ users using WASM cryptography, reducing wallet sync time 60% via batched UTXO filtering
- Engineered Firestore sync for stateless bots, cutting tx failures 99% across 1K+ weekly transfers via pre-flight validation

PROJECTS

Minimal LLM Inference Engine | Rust, SIMD, GGUF, Llama | [GitHub](#)

Jan 2026 – Present

- Engineered a zero-copy Rust LLM engine with memory-mapped GGUF parsing, KV-cache, and GQA; 923 tests
- Developed INT8 quantization and parallel SIMD GEMM kernels, yielding 21x speedup, 4x memory savings

Distributed KV Database | Java, gRPC, Kubernetes, Prometheus, Grafana | [GitHub](#)

Jul 2025 – Present

- Developed a distributed KV database with a custom LSM-tree storage engine and dual Raft/quorum consensus via gRPC
- Built Prometheus/Grafana monitoring and a Jepsen-style linearizability checker, cutting p99 latency via hedged requests

TECHNICAL SKILLS

Languages & Frameworks: Rust, Java, Python, Go, SQL, C++, TypeScript, React, FastAPI

Distributed Systems: Spark, Kafka, gRPC, Redis, RabbitMQ, Apache Arrow

ML/AI: PyTorch, TensorFlow, JAX, ONNX, GGUF

Databases: PostgreSQL, ClickHouse, Elasticsearch, DynamoDB, MongoDB

Cloud & Infrastructure: Kubernetes, AWS, GCP, Docker, Terraform, CI/CD, Git, Prometheus, Grafana

Certificates: [Google Cloud Certified – Professional Cloud Architect](#)

LEADERSHIP & AWARDS

Research Apprenticeship Award | Northeastern University

Fall 2025

President / Co-Director – NU App Lab | Northeastern University

Fall 2025 – Present